

Technical Architecture

1. Frontend Layer

The Frontend serves as the client-side application where users and administrators interact with the system. It provides modules such as User Authentication, Product Listing, Product Details, Search & Filter, Shopping Cart, Checkout, User Profile, Order History, Admin Dashboard, Product Management, Order Management, and User Management.

Technologies: React.js/Angular/Vue.js, HTML5, CSS3, JavaScript/TypeScript, Axios/Fetch API.

2. Backend Layer

The Backend exposes REST APIs and implements business logic including Authentication, User, Product, Cart, Order APIs, Payment Integration, Admin Authentication, and Admin Dashboard APIs. Responsibilities include JWT authentication, password encryption, CRUD operations, order processing, inventory management, authorization, validation, and logging.

Technologies: Node.js, Express.js, JWT, Bcrypt, REST APIs.

3. Database Layer

Stores Users, Products, Cart, and Orders collections. Responsible for persistent storage, inventory, carts, orders, and data integrity.

Technology: MongoDB.

4. Communication Flow

1. User interacts with Frontend.
2. Frontend sends HTTP requests to Backend.
3. Backend authenticates and validates requests.
4. Backend executes business logic.
5. Backend accesses Database.
6. Database returns data.
7. Backend responds with JSON.
8. Frontend updates the UI.

Architecture Summary

Frontend: UI and API communication.

Backend: Business logic and REST APIs.

Database: Persistent data storage.

This is a three-tier architecture consisting of Presentation, Application, and Data layers.